Chapter 8: Instruction Encoding and ISA (Live)

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N4-02c-92

Learning Objectives

- Describe the instruction format of ARM instruction-set architecture (ISA)

- Describe differences between fixed and variable-length ISA

Overall Instruction Format

- Arithmetic & Logic instructions
  - ADD, ADC, SUB, SBC, RSB, AND, EOR, RSC, ORR, BIC

XXXCCS Rd, Rs1, Rs2 , {shft #}

31 28		24 21		19 16	15 12	11 7	6 5		3 0
Condition	000	Inst. type	S	Rs1	Rd	Shift size	shft	0	Rs2

XXXCCS Rd, Rs1, Rs2 , {shft Rshift}

31 28		24 21		19 16	15 12	11	7	6 5		3 0
Condition	000	Inst. type	S	Rs1	Rd	Rshift	0	shft	1	Rs2

XXXCCS Rd, Rs1, #immediate (rotated)

31 28		24 21		19 16	15 12	11 8	7 0
Condition	001	Inst. type	S	Rs1	Rd	#rot	8-bit immediate

Overall Instruction Format

- Arithmetic & Logic instructions
  - ADD, ADC, SUB, SBC, RSB, AND, EOR, RSC, ORR, BIC

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31 28		24 21		19 16	15 12	11 7	6 5		3 0
Condition	000	Inst. type	S	Rs1	Rd	Shift size	shft	0	Rs2

ADD R0,R1,R2,LSL #5

31 28		24 21		19 16	15 12	11 7	6 5		3 0
1110	000	0100	0	0001	0000	00101	00	0	0010

ADD R0,R1,R2

31 28		24 21		19 16	15 12	11 7	6 5		3 0
1110	000	0100	0	0001	0000	00000	00	0	0010

Q1 which of the following hexadecimal encoding of ADD R2,R1,R3

1. 0xE0812003
2. 0xF1021003
3. 0XE0831200
4. 0x00012003

XXXCCS Rd, Rs1, Rs2 , {shft #}

31 28 24 21 19 16 15 12 11 7 6 5 3 0

Condition	000	Inst. type	S	Rs1	Rd	Shift size	shft	0	Rs2

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XXXCCS Rd, Rs1, #immediate (rotated)

31 28 24 21 19 16 15 12 11 8 7 0

Condition	001	Inst. type	S	Rs1	Rd	#rot	8-bit immediate

Code	Condition	Flags	Meaning
0000	**EQ**	Z = 1	Equal
0001	**NE**	Z = 0	Not equal
0010	**CS or HS**	C = 1	Higher or same, unsigned
0011	**CC or LO**	C = 0	Lower, unsigned
0100	**MI**	N = 1	Negative
0101	**PL**	N = 0	Positive or zero
0110	**VS**	V = 1	Overflow
0111	**VC**	V = 0	No overflow
1000	**HI**	C = 1 and Z = 0	Higher, unsigned
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1011	**LT**	N != V	Less than, signed
1100	**GT**	Z = 0 and N = V	Greater than, signed
1101	**LE**	Z = 1 and N != V	Less than or equal, signed
1110	**AL**		Always.
1111	**NV**		Reserved (unused)

Code	Instruction
0000	**AND**
0001	**EOR**
0010	**SUB**
0011	**RSB**
0100	**ADD**
0101	**ADC**
0110	**SBC**
0111	**RSC**

Code	Instruction
1000	**TST**
1001	**TEQ**
1010	**CMP**
1011	**CMN**
1100	**ORR**
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1110	**BIC**
1111	**MVN**

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Q2 which of the following hexadecimal encoding of RSBEQS R2,R1,#5

1. 0xF1012005
2. 0x02712005
3. 0X01321005
4. 0x01212005

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31–28	27–25	24–21	20	19–16	15–12	11–7	6–5	4	3–0
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Condition	000	Inst. type	S	Rs1	Rd	Rshift	0	shft	1	Rs2

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Overall instruction format

- MOV instructions
  - `MOV, MVN`

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Shift/Rotate instructions

- Instructions are encoded as MOV instructions

  E.g. LSL R1, R2, #5 ≡ MOV R1, R2, LSL #5

- Specify the shift type in bits 5 and 6

Code	Shift type
00	LSL
01	LSR
10	ASR
11	ROR/RRX

XXXCCS Rd, Rs , {shft #}

31–28	27–25	24–21	20	19–16	15–12	11–7	6–5	4	3–0
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Q3 which of the following hexadecimal encoding of MOVLE R7,R4

1. 0xD0D07400
2. 0xD0D07004
3. 0XD1A07004
4. 0xD1A07400

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XXXCCS Rd, Rs , {shft #}

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1000	**TST**
1001	**TEQ**
1010	**CMP**
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1100	**ORR**
1101	**MOV**
1110	**BIC**
1111	**MVN**

Branch Instructions

- For conditional branch and branch with link

B(L)CC Offset

31	28	23	0
Condition	101	L	Offset

- 26 bits offset shifted to the right by 2 (branching to word addresses) ➔ calculated by the assembler

- Branch range: ±32 MBytes

Load/Store Instructions

- Load/Store word bye

LDR/STR{B}CC Rs,[RD,offset pre or post]W

31–28	27–26	25	24–21	20	19–16	15–12	11–7	6–5	4	3–0
Condition	01	0	P U B W	L	Rs	Rd	Shift size	shft	0	Rs

31–28	27–26	25	24–21	20	19–16	15–12	11–0
Condition	01	1	P U B W	L	Rs	Rd	Immediate value

- L bit determines load (1) or store (0)
- P bit determines indexing pre (1) or post (0)
- U determines offset direction add (1) or subtract (0)offset
- B determines byte (1) or word (0) access
- W is for write back (!) of the offset

Load/Store Instructions

- Load/Store Multiple instructions

```
LDM/STMFXCC   RsW,{register list}
```

31 28		24 21		19 16	15 12 11 0
Condition	100	P U ^ W	L	Rs	Register list

- L bit determines load (1) or store (0)
- P bit determines indexing before (1) or after (
- U determines offset direction add (1) or subtr
- ^ is “don’t care”
- W is for write back (!) after operation

PU	Instruction
10	STMFD
01	LDMFD
11	STMFA
00	LDMFA

Load/Store Instructions

- Load/Store Multiple instructions

`STMFD SP!,{R0-R6}`

31 28	24	21		19 16	15															0
1110	100	10^1	0	1101 (SP)	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1
					PC	LR	SP	R12	R11	R10	R9	R8	R7	R6	R5	R4	R3	R2	R1	R0

- L bit determines load (1) or store <u>(0)</u>
- P bit determines indexing before <u>(1)</u> or after (0)
- U determines offset direction add (1) or subtract <u>(0)</u>
- ^ is “don’t care”
- W is for write back (!) after operation <u>(1)</u>

Q4 If the registers list field is encoded with 0X40F0, what register list this represents?

31–28	27–25	24–21	20	19–16	15–0
Condition	100	P U ^ W	L	Rs	Register list

1. {R4,R15}

2. {R4-R7,LR}

3. {R1,R8-R11}

4. Don't know!

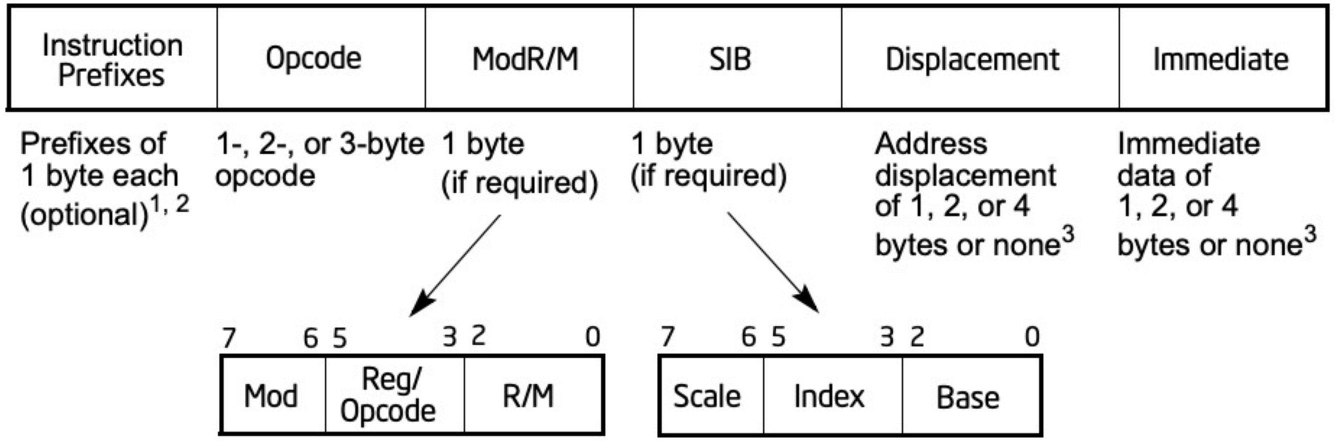
Answer is 2

Fixed vs. Variable-Length instruction

- ARM is fixed-length
  - Operands had to be registers or immediate values
- Variable length instructions
  - Intel 80x86: Instructions vary from 1 to 17 bytes long.
  - Digital VAX: Instructions vary from 1 to 54 bytes long.
  - Require multi-step fetch and decode.
  - Allow for a more flexible (but complex) and compact instruction set.

Variable-Length: Intel Instruction Format

- Complex instructions = complex hardware
- Higher variety for smaller code and faster execution

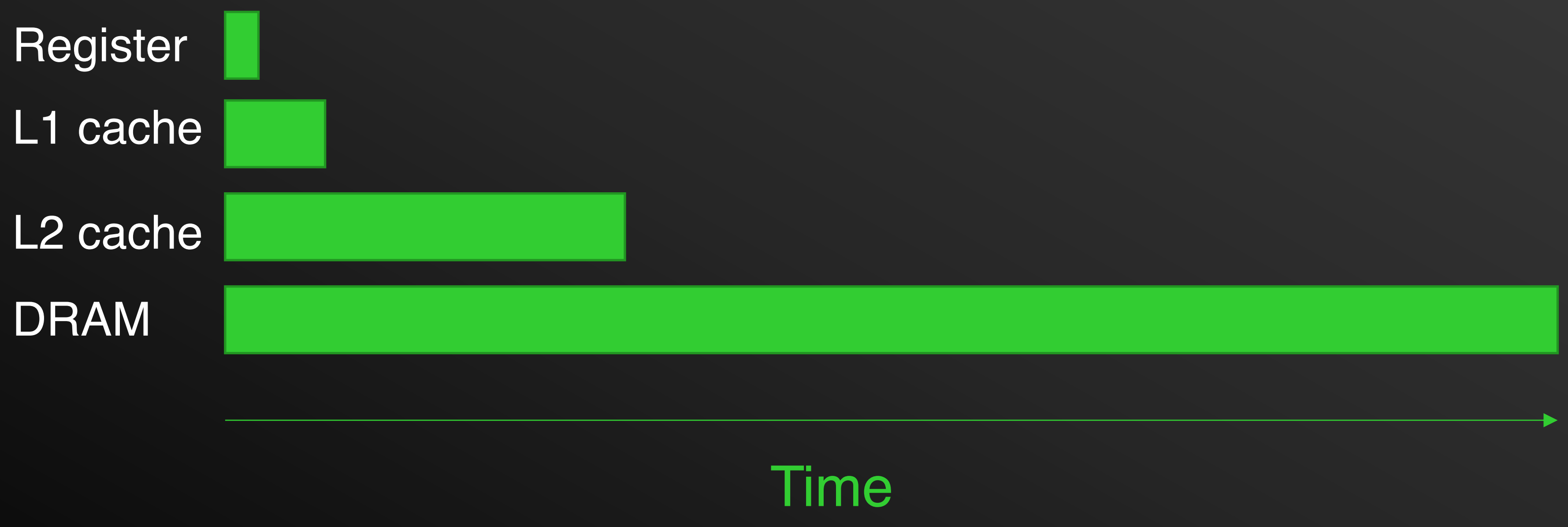


1. The REX prefix is optional, but if used must be immediately before the opcode; see Section 2.2.1, "REX Prefixes" for additional information.
2. For VEX encoding information, see Section 2.3, "Intel® Advanced Vector Extensions (Intel® AVX)".
3. Some rare instructions can take an 8B immediate or 8B displacement.

Figure 2-1. Intel 64 and IA-32 Architectures Instruction Format

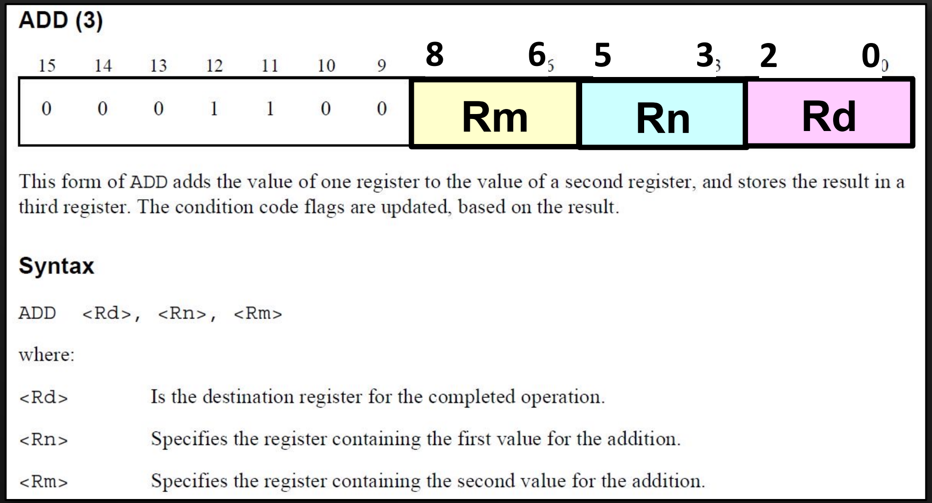
Using Registers

- Operations using registers are **faster** than those involving memory.
  - Data transfer between registers and ALU is faster due to it physically proximity.
  - To speed up code execution, keep as much of your computations within **registers**.



Implication of Register Count

- Implementing many registers within the CPU will incurs increasing overheads.
    - They take up precious **space** in the silicon die.
    - Connections to various busses increase the **routing** and **multiplexing complexity**.
    - More registers increases the **operand size** during instruction encoding.



ADD (3)

15	14	13	12	11	10	9	8 – 6	5 – 3	2 – 0
0	0	0	1	1	0	0	Rm	Rn	Rd

This form of ADD adds the value of one register to the value of a second register, and stores the result in a third register. The condition code flags are updated, based on the result.

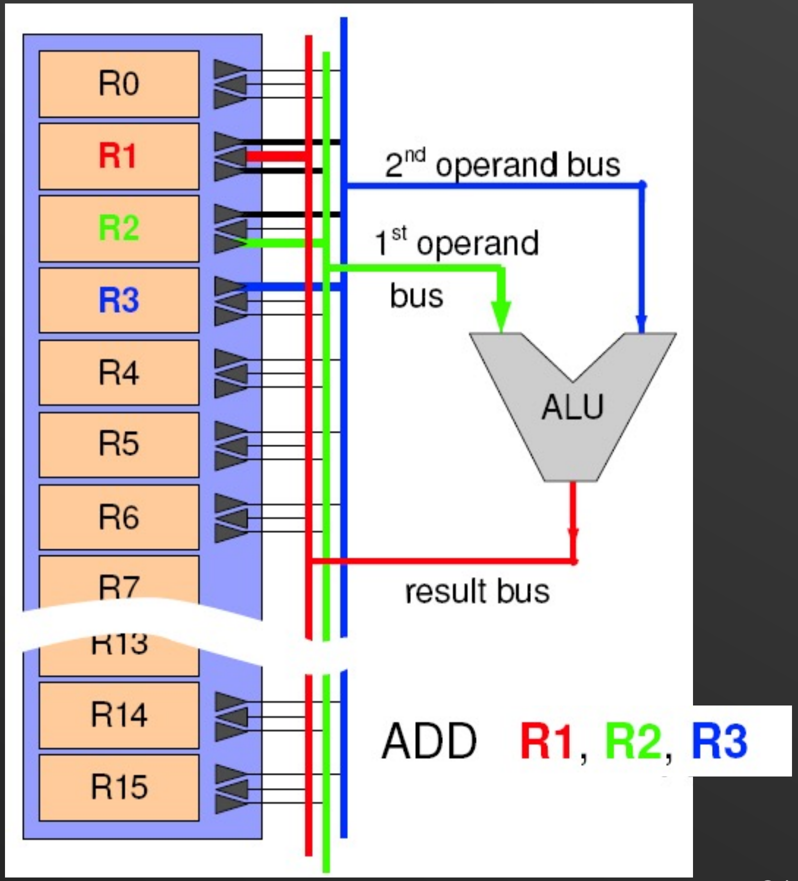
Syntax

`ADD <Rd>, <Rn>, <Rm>`

where:

<Rd>	Is the destination register for the completed operation.
<Rn>	Specifies the register containing the first value for the addition.
<Rm>	Specifies the register containing the second value for the addition.

Instruction Encoding for ADD in Thumb-2



Orthogonality of ISA

- In a truly orthogonal ISA, every instruction is able to use every available addressing modes.
- A truly orthogonal ISA does not restrict certain instruction from using only specific registers.
- Difficult to achieve complete orthogonality due to the **large number of bit patterns** required to express all combinations of operations and addressing modes.
- Increase instruction length will increases the memory size required by the program.

Summary - Why is Good Instruction Set Architecture (ISA) Design Important?

- Good ISA design can yield benefits such as:
  - Increases the execution performance of the processor.
  - Increases the code density (i.e. how much memory a piece of code will occupy).
  - Reduces the power consumption of the CPU.
  - Reduces manufacturing cost of processor.
  - Easy for programmers to write efficient programs.
  - Efficient mapping of high-level programming requirements into low-level instruction sets.

Why Should I Care About Processor Designs and ISA?

- Designing a processor (or a co-processor) is no longer a hardware company job

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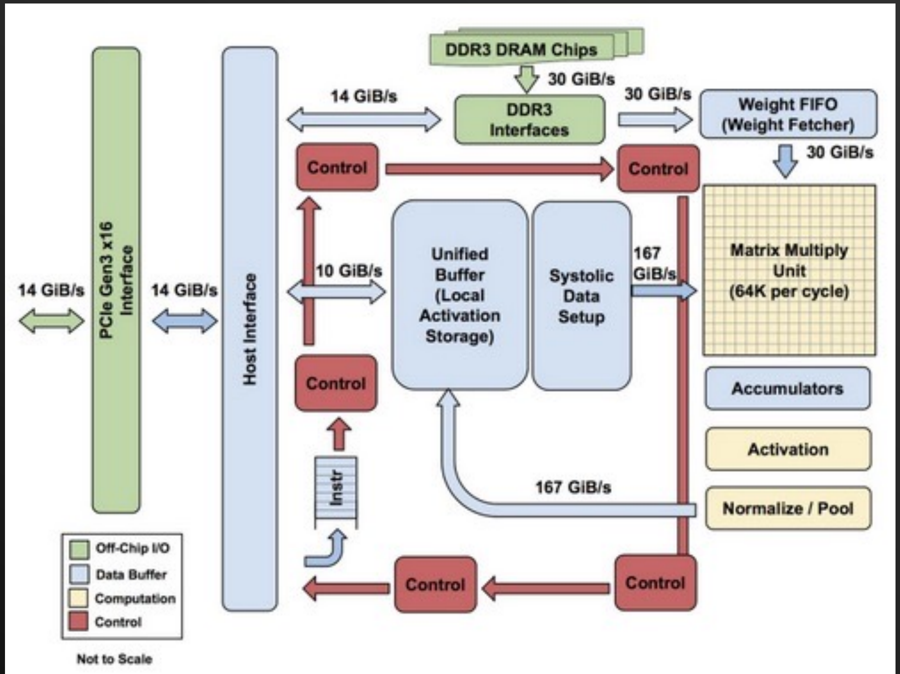
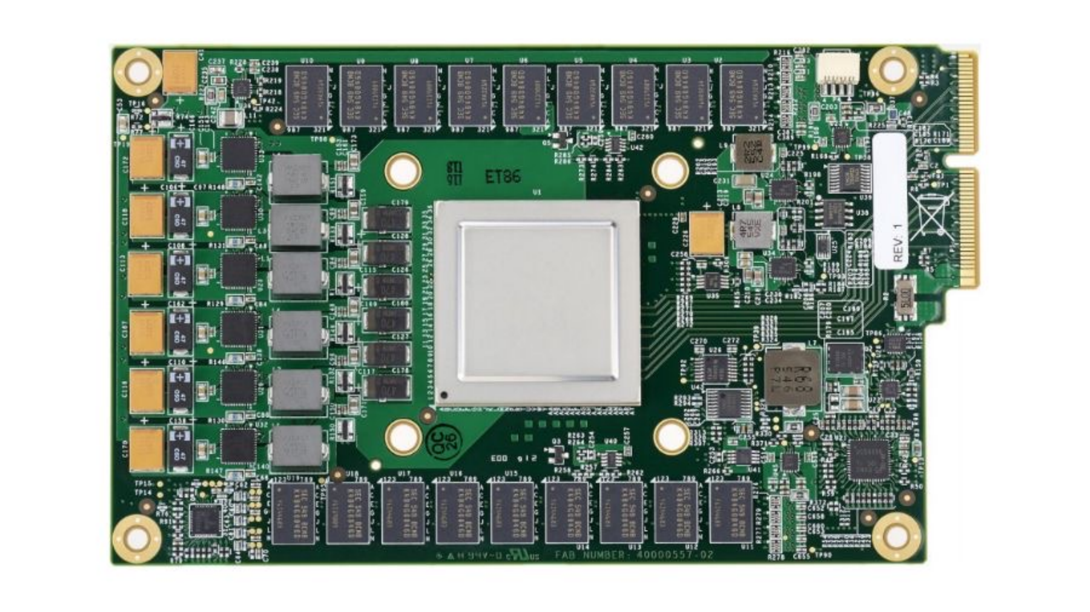
The Google TPU

- Tensor Processing Unit
- Designed by Engineers in Google
- Tailored to make AI applications run faster





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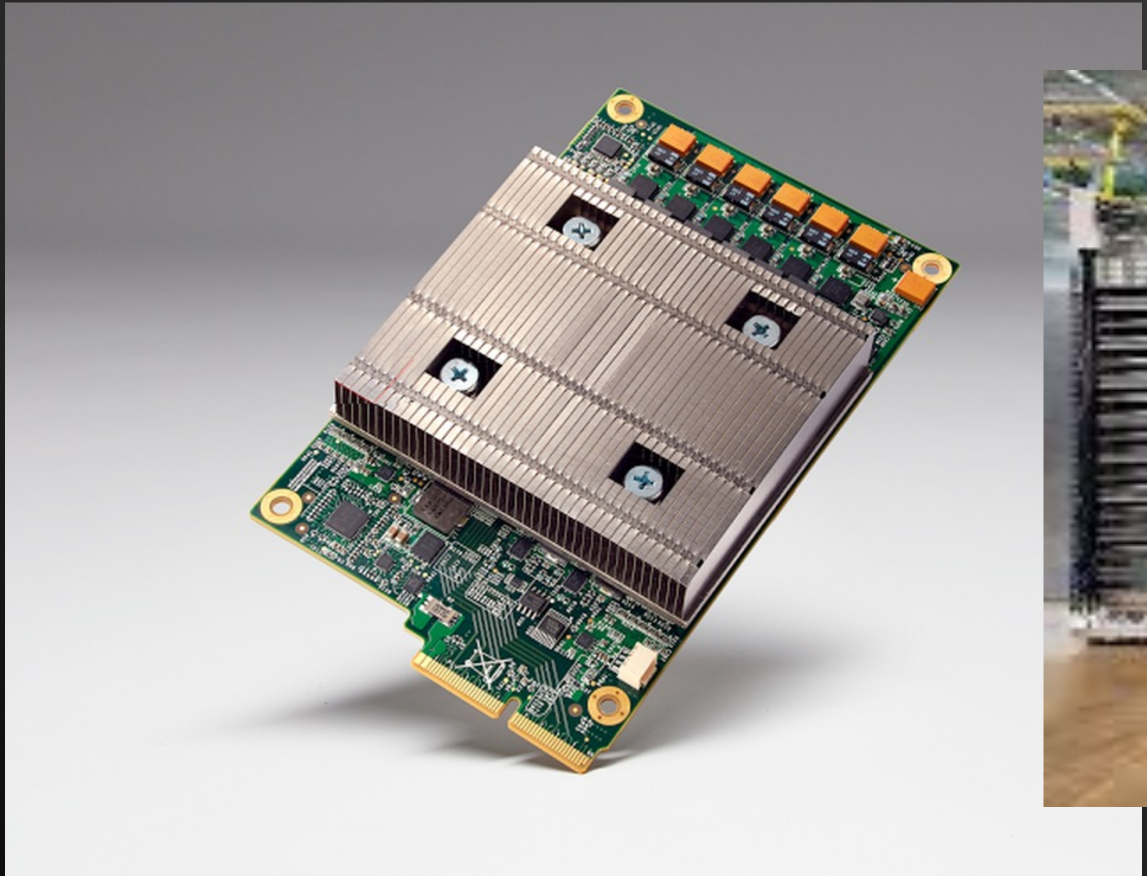
- Design by Engineers in Google to make AI applications run faster



Summary

- ARM is fixed-length load/store architecture
  - Operands reside in registers
- Other architectures can use variable-length format
  - E.g, Intel architecture
- Use of registers is key towards higher performance